Project Synopsis Template

1. Title

"Startup Success Prediction using Machine Learning"

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
M. Lahari	MRECW	CSE-DS	7207646839	laharimallela2324@gmail.com	Team Lead
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Program Track and Domain

Program Track (Engg-HUB / Engg-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Engg-Spoke	Machine learning

Guide Details

Guide Name: R. Manoj Kumar(SME)

2. Problem Statement

The rapid growth of startups has increased uncertainty in identifying high-potential ventures. Investors and entrepreneurs often depend on intuition and limited analysis, leading to high failure rates.

Traditional evaluation methods are time-consuming, subjective, and inefficient for large datasets. There is a need for a data-driven system to accurately predict startup success. Such a system should analyze factors like funding, market conditions, founder experience, and business metrics.The objective is to develop a machine learning model using historical startup data.This model should identify key success indicators and provide reliable predictions.The startup ecosystem is dynamic, making continuous monitoring and evaluation challenging. Existing approaches lack scalability and fail to capture complex nonlinear relationships. Therefore, a scalable and adaptive ML-based solution is required to support proactive, evidence-based decision-making.

3. Objectives

- To develop an accurate machine learning model for predicting startup success.
- To analyze historical startup data and identify key success factors. • To perform effective data preprocessing and feature engineering to improve performance.
- To compare multiple ML algorithms and select the best-performing model.
- To evaluate the model using Accuracy, Precision, Recall, and F1-Score. • To build a scalable and interpretable decision-support system for early risk assessment.

4. Methodology / Implementation Plan

The methodology begins with collecting historical startup data from reliable public datasets and sources. The data is preprocessed through cleaning, handling missing values, and normalization to ensure quality and consistency. Feature engineering is applied to extract important attributes such as funding, founder experience, and market trends, followed by feature selection to identify the most influential factors. The dataset is then divided into training and testing sets for model development. Multiple machine learning models, including Logistic Regression, Random Forest, and SVM, are trained and compared. Finally, the models are evaluated using metrics like Accuracy, Precision, Recall, F1-Score, and ROC-AUC, and the best-performing model is selected for deployment as a predictive system.

Tools and Technologies Used:

1. Python Programming Language
2. • PyTorch Framework
3. • Random Forest Classifier (Machine Learning Model)
4. • Scikit-learn Library
5. • Jupyter Notebook / Google Colab Environment

5. Resources and Timeline

A. Hardware & Software Requirements

Laptop/PC with Intel i5 / AMD Ryzen 5 Processor
● Minimum 8GB RAM (16GB Recommended) ●
Windows / Linux / Ubuntu Operating System
● Python 3.10 or Higher
● PyTorch Framework
● NumPy, OpenCV, Matplotlib Libraries
● Jupyter Notebook / Google Colab Environment
● Minimum 10GB Free Storage Space

B. Dataset

Source: Kaggle Dataset Repository

Dataset Name: Startup Succes Prediction Dataset

The dataset contains historical startup records labeled as successful or unsuccessful. It includes features such as funding amount, founder experience, industry type, market size, and financial indicators, which are used for supervised training and evaluation of machine learning models to predict startup success.

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C. Project Timeline

Phase	Task	Tentative Completion
Phase 1	Problem Analysis & Dataset Selection	Week 1

Phase 2 Data Preprocessing & Preparation	Week 2
Phase 3 Model Development using MobileNetV2	Week 3
Phase 4 Model Training & Evaluation	Week 4
Phase 5 Streamlit Deployment & Testing	Week 5
Phase 6 Documentation & Final Presentation	Week 6

6. Expected Outcomes / Conclusion

The project aims to develop a functional machine learning-based system capable of accurately predicting the success or failure of startups using historical data. By integrating classification models such as Logistic Regression, Random Forest, SVM, and ANN with a structured preprocessing and evaluation pipeline, the system provides a reliable and datadriven decision support solution. The expected outcome includes a trained and optimized predictive model, performance evaluation results using metrics like Accuracy and F1-Score, and a deployable prototype that demonstrates practical application in investment analysis and strategic business planning within the startup ecosystem.

7. References

IEEE Access – Predicting Startup Success Using Machine Learning Algorithms (Brown & Smith, 2022)

- Journal of Business Analytics – Machine Learning Approaches for Startup Performance Prediction (Lee & Park, 2023)
- International Journal of Data Science and Analytics – Feature Engineering Techniques for Startup Success Forecasting (Kumar & Gupta, 2024)
- Kaggle – Startup Success Dataset and Analysis (2023)
- arXiv – A Hybrid Machine Learning Framework for Predicting Startup Outcomes (Zhao et al., 2023)